AMTEC
AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 109

LAKAS KULIGLIG Recirculating Grain Dryer, Model: RMD-LK 66

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LAKAS KULIGLIG RMD-LK 66 RECIRCULATING GRAIN DRYER

Test Report No.	:	2013 - 109
Date of Test	:	September 14-15, 2013
Test Valid Until	:	September 15, 2018

Brand	:	LAKAS KULIGLIG
Model	:	RMD-LK 66
Type	:	Batch, recirculating with indirect-fired biomass-fed furnace
Manufacturer	:	P.I. FARM PRODUCTS, INC. 35 Alarcon St., Sitio Malabo Maysan, Valenzuela City
Test Requested by:	:	Manufacturer
Machine Classification	:	Commercial

SUMMARY

The LAKAS KULIGLLIG RMD-LK 66 Recirculating Grain Dryer with indirect-fired biomass-fed furnace was tested for performance using PALAY as test material. The Dryer had a load capacity of six tons. It had a Drying Capacity of 400.0 kg/h from an initial grain Moisture Content of 29.1%$_{w.b.}$ to a Final Moisture Content of 13.9%$_{w.b.}$ with an average Moisture Reduction per Hour of 1.02%. The actual Drying Time for the palay with an initial weight of 6000 kg was 15.0 hours. The average Drying Air Temperature during the test was 60.8^{o}C. The Fuel

Consumption of the indirect-fired biomass furnace using rice hull as fuel was 24.7 kg/h while the Total Power Consumption of the electric motors which drove the components of the dryer was 7.30 kW with Total Load Current of 42.1 amperes and Line Voltage of 221 volts.

The Heating System Efficiency of the dryer was 53.7%. Heat Utilization was 2600 kJ per kg moisture removed with a corresponding Drying Efficiency of 90.5%. The dryer had a Drying System Efficiency of 48.5%. Based on the moisture content of samples taken at the end of drying Moisture Content Variation was ± 0.24 % while the Moisture Gradient was 1.10%.

PURPOSE AND SCOPE OF TEST

The LAKAS KULIGLIG RMD-LK 66 Batch, Recirculating Grain Dryer coupled to an indirect-fired biomass-fed furnace was tested for drying PALAY to determine its operating performance. The test was conducted to verify and assess the following:

1.0 Drying System Performance
- 1.1 Drying Capacity
- 1.2 Moisture Reduction per Hour
- 1.3 Heat Utilization
- 1.4 Drying Efficiency
- 1.5 Drying System Efficiency

2.0 Heating System
- 2.1 Heating System Efficiency
- 2.2 Furnace Fuel Consumption Rate (using rice hull)

3.0 Quality of Dried Grains
- 3.1 Cracked Grain
- 3.2 Milling Quality
- 3.3 Hulled/Damaged Grain
- 3.4 Moisture Content Variation

PLACE AND DATE OF TEST

The LAKAS KULIGLIG RMD-LK 66 Batch, Recirculating Grain Dryer was tested at the plant site of P. I. Farm Products, Inc., in Sitio Malabo, Maysan, Valenzuela City on September 14-15, 2013.

METHODS OF TEST

The LAKAS KULIGLIG RMD-LK 66 Batch, Recirculating Grain Dryer was tested based on the Philippine Agricultural Engineering Standard "PAES 202:2000 Agricultural Machinery – Heated-Air Mechanical Grain Dryer - Methods of Test".

OPERATOR'S MANUAL

No operator's manual or brochure was submitted.

DESCRIPTION OF THE GRAIN DRYER

Figure 1 shows the LAKAS KULIGLIG RMD-LK 66 Batch, Recirculating Grain Dryer coupled to an indirect-fired biomass furnace. It was basically a continuous flow type dryer utilizing the LSU (Louisiana State University) drying system operated as batch-recirculating. It consisted of the following main components: grain loading system, drying system, control panel, biomass furnace and bagging bin.

LOADING SYSTEM

The loading system consisted of a loading hopper (**Figure 2**) and a bucket elevator (**Figure 3**). It was used for loading grains to the drying columns. The loading hopper with a platform was located on the lower portion of the elevator alongside the drying column. The elevator was a belt-bucket centrifugal discharge type.

DRYING SYSTEM

The drying system consisted of a drying chamber, a hot-air fan and a grain unloading system.

Drying Column (**Figure 4**)

This was where the wet grains were being dried. Its configuration was based on the LSU design. The chamber was divided into three modules, namely: the buffer module (**Figure 4a**); drying module (**Figure 4b**); and, the unloader module (**Figure 4c**). The buffer module served as the holding bin for the wet grain. It was used mainly for sealing hot air in the drying chamber. The drying module was where hot air passed and mixed the grains in a cross-flow direction. It was made of alternating rows of intake and exhaust ducts. The cooling/unloader module was at the bottom of the drying column and provided with a discharging system to regulate the discharge flow. The output of the unloader was conveyed to the elevator thru a belt conveyor.

Hot air Fan. (**Figure 5**)

The dryer was provided with axial-flow type hot-air fan. This was driven by a 3.73 kW electric motor through belt and pulley. This was installed on one side of the dryer opposite the air duct. The fan pulled the hot air that passed through the grain. The exhaust air was conveyed through an air duct wherein the dusts and impurities were deposited in the dust collector **(Figure 6)**

CONTROL PANEL (**Figure 7**)

The dryer control system was housed in a rectangular box compartment mounted near the elevator. It included the switches and thermostat control panel of the dryer.

BIOMASS FURNACE (**Figure 8**)

The dryer was coupled to an indirect-fired furnace and which used either corn cobs or rice hull as fuel. The combustion chamber was lined with insulating firebricks while the heat exchangers (**Figure 9**) were made of 63 pieces of mild steel pipe installed horizontally. Fuel feeding was done mechanically by means of a plunger-bed type (**Figure 10**) feeding mechanism installed right at the bottom outlet of the hopper. A centrifugal fan (**Figure 11**) provided primary air underneath the burning chamber through holes in the brick floor. Ash fell beneath the burning chamber to the ash discharge conveyor **(Figure 12)**. The gas was sucked by a chimney through the heat exchangers. Ambient air was heated when it entered at one end of the heat exchangers and exit through the other end leading to the dryer.

Bagging Bin **(Figure 13)**

This was used as temporary storage of dried paddy prior to bagging.

Figure 1. LAKAS KULIGLIG Batch,
Recirculating Grain Dryer
Model: RMD-LK 66

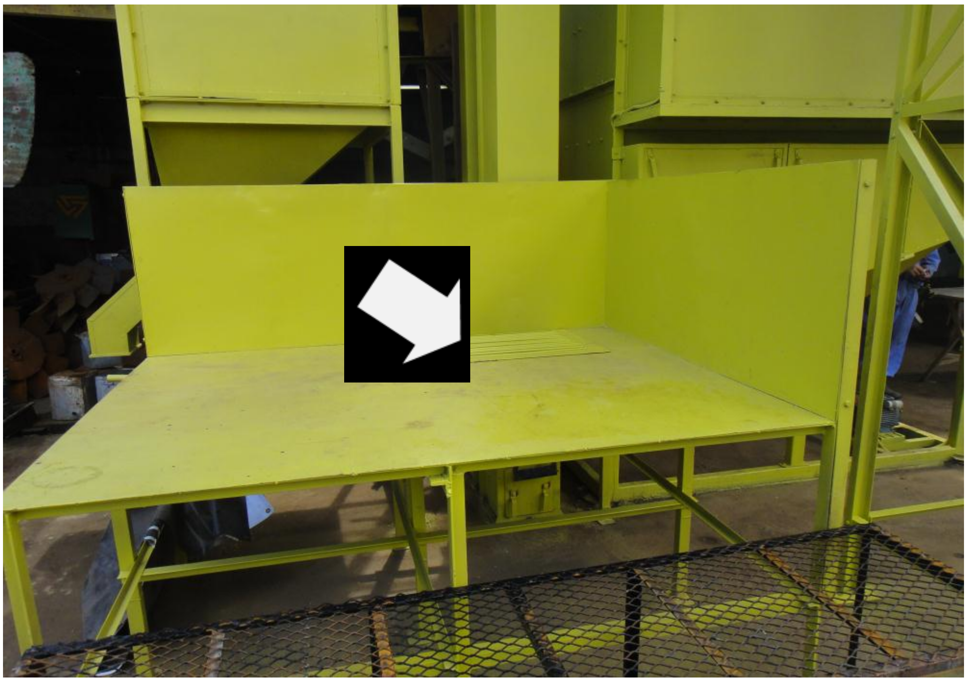

Figure 2. Loading hopper

Figure 3. Elevator

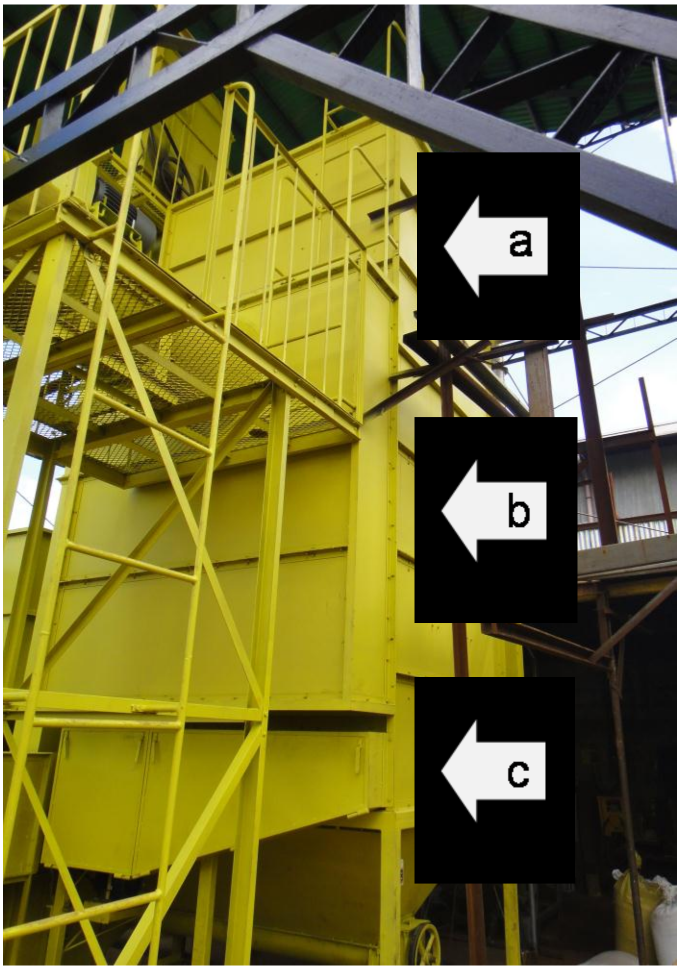

Figure 4. Drying column
a) Buffer module
b) Drying module
c) Unloader module

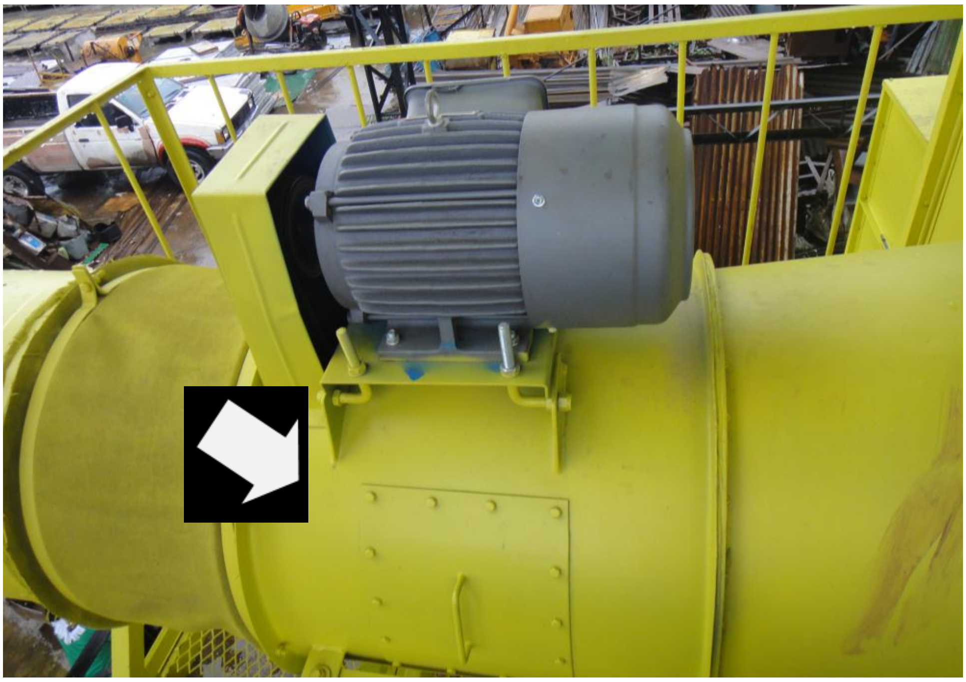

Figure 5. Hot air fan

Figure 6. Dust Collector

Figure 7. Control Panel

Figure 8. Indirect-Fired Biomass Furnace

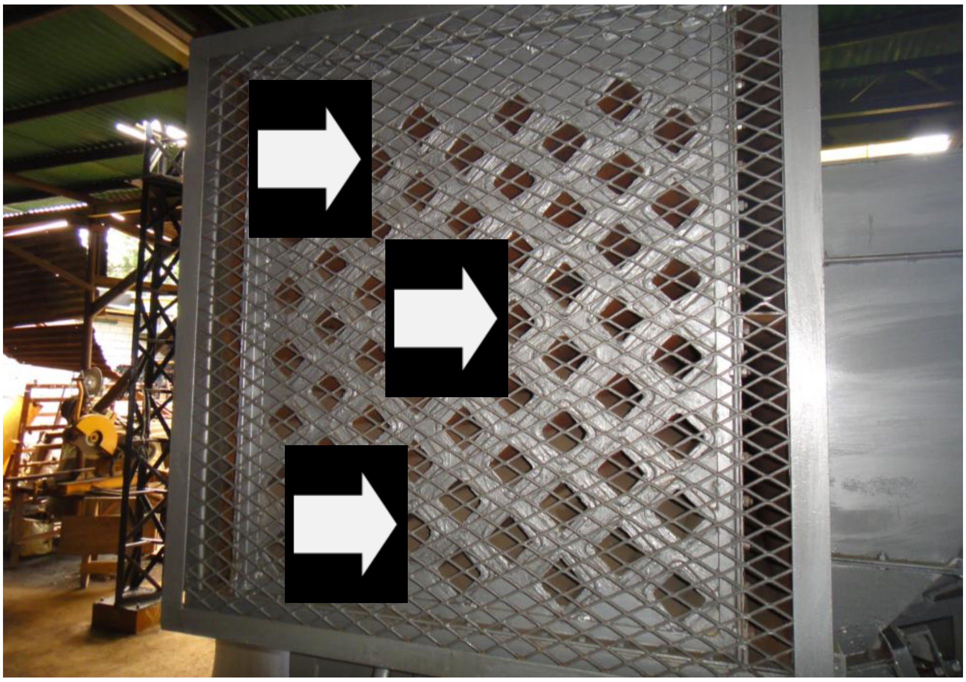

Figure 9. Heat Exchangers

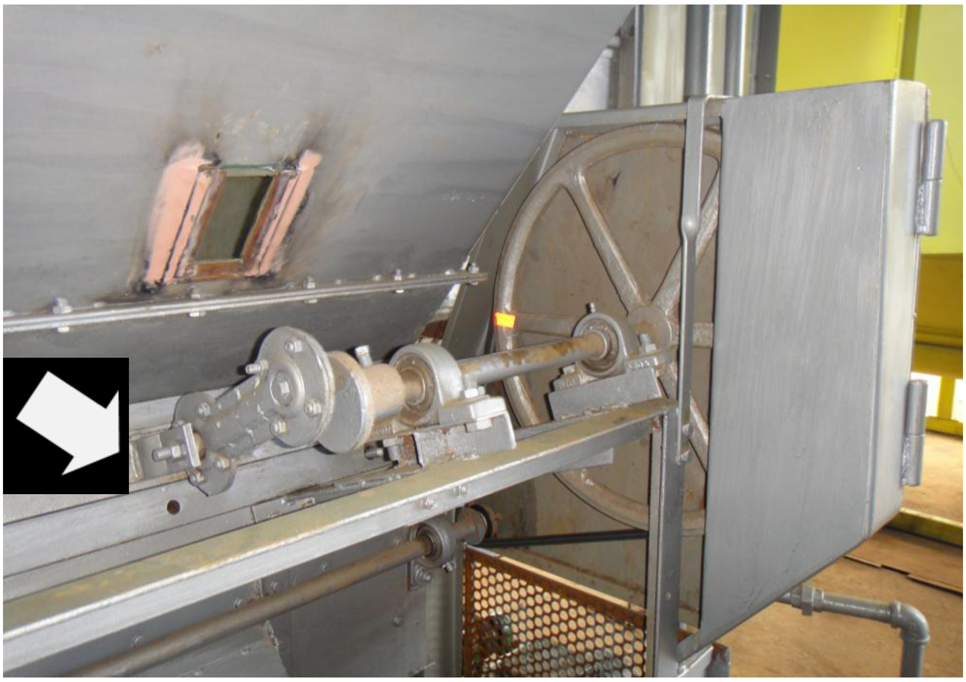

Figure 10. Plunger-bed type feeding mechanism

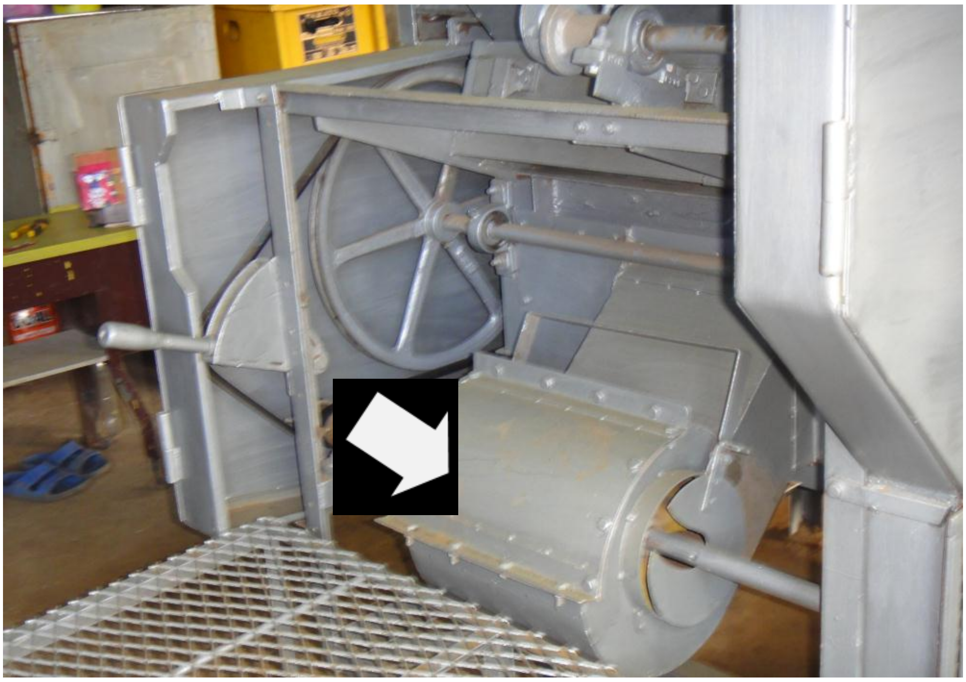

Figure 11. Fan for supply of combustion air

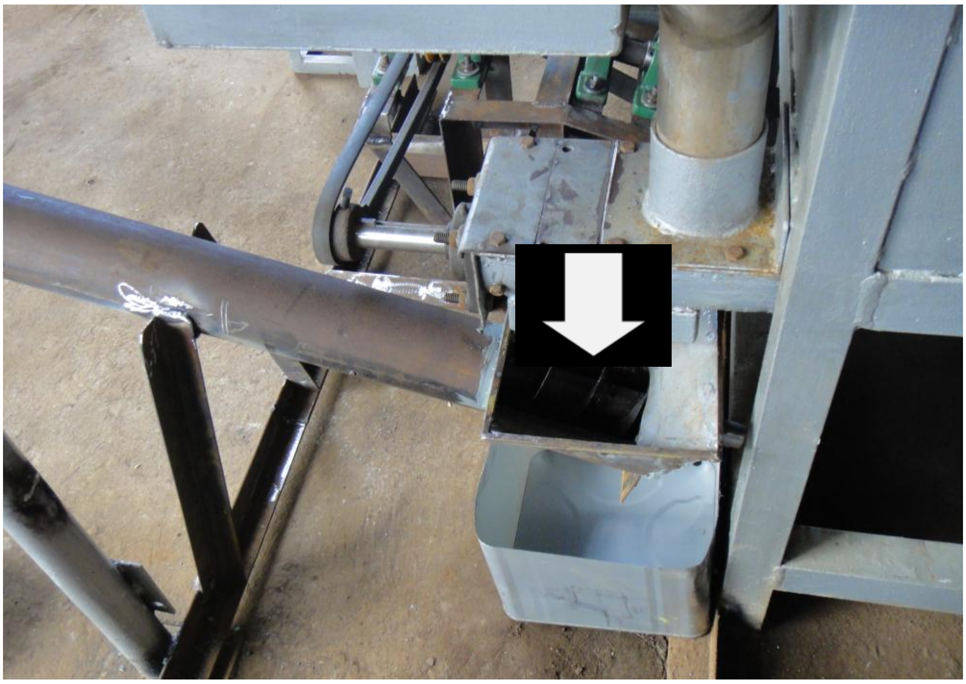

Figure 12. Ash Discharge Conveyor

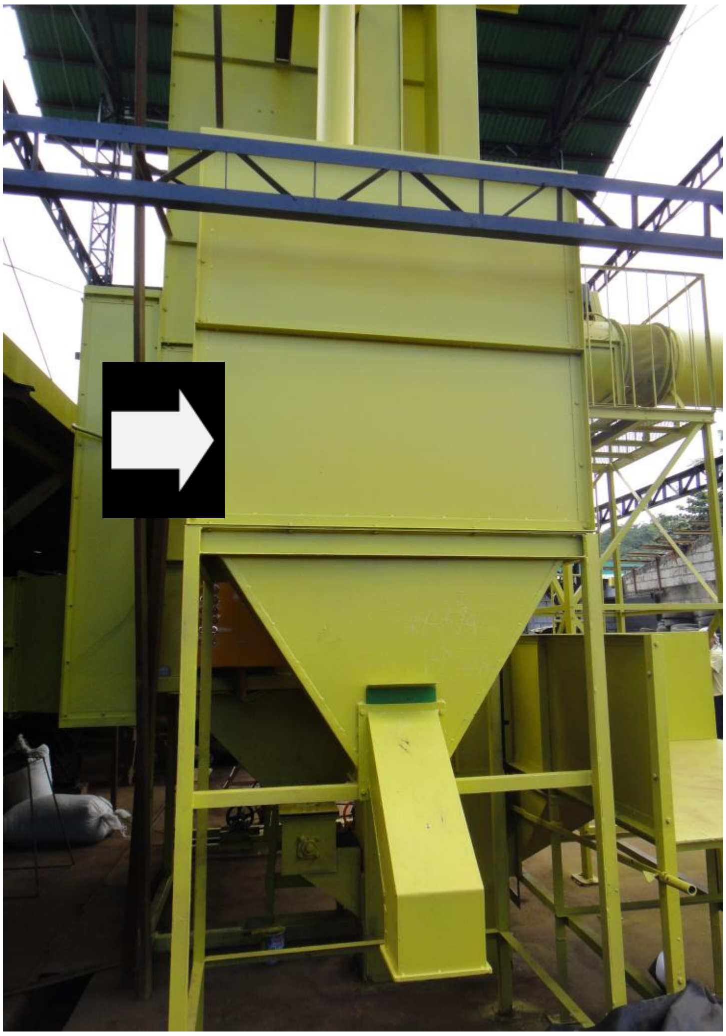

Figure 13. Bagging Bin

RESULTS AND DISCUSSION

The LAKAS KULIGLIG RMD-LK 66 Batch, Recirculating Dryer coupled to a biomass-fed furnace was tested by filling up the drying chamber with PALAY with average moisture content of 29.1 $\%_{W.B}$ at its capacity. The furnace using rice hull as fuel was fired and when the drying air temperature became stable, the dryer was started. The grains inside the drying chamber moved downward as the bottom layers were recirculated to the top of the drying chamber. Drying operation was stopped when the moisture content of the grains at the discharge of the drying section reached about 14.0%.

Results of performance tests are shown in **Table 1.**

Drying Capacity

The drying capacity of the dryer was the amount of fresh palay dried per unit of time, from initial moisture content until the desired moisture content was attained. The dryer had a Drying Capacity of 400.0 kg/h from an initial moisture content of 29.1$\%_{W.B.}$ to a final moisture content of 13.9$\%_{W.B.}$.

Moisture Content Reduction

There was an average Moisture Content Reduction during the drying process of 1.02% per hour.

Moisture Content Variation

Based on the moisture content of samples taken from the outlet chute of the dryer, Moisture Content Variation at the termination of drying was ±0.24%.

Heating System Efficiency

Heating system efficiency was the ratio of the amount of heat supplied for drying to the total amount of heat available from the fuel, expressed in percent. The dryer had Heating System Efficiency of 53.7%.

Furnace's Fuel Consumption

To determine the fuel consumption of the indirect-fired biomass furnace, the total amount of rice hull consumed during the test was divided by the total time of operation. The Fuel Consumption of the furnace was 24.7 kg/h of rice hull.

Heat Utilization

Heat utilization referred to the total amount of heat utilized to vaporize moisture from the grain. Results showed that the dryer had a Heat Utilization of 2600 kJ/kg of water removed.

Drying Efficiency

Drying efficiency was the ratio of the total heat utilized to vaporize moisture in the grain to the amount of heat added to the drying air expressed in percent. Results showed that the dryer had a Drying Efficiency of 90.5%

Drying System Efficiency

Drying system efficiency was the ratio of the total heat utilized for drying to the heat available in the fuel expressed in percent. Results showed that the dryer had a Drying System Efficiency of 48.5%.

Power Consumption

The average total Power Consumption of the dryer during operation was 7.30 kW with Load Current of 42.1 amperes and Line Voltage of 221 volts.

Labor Requirement

One person **(Figure 14)** was needed to operate the dryer i.e. to replenish fuel to the furnace, during the drying operation.

Observations

The dryer operated smoothly and no breakdown or failure of components occurred during the test.

1. The elevator's hopper was provided with grill cover **(Figure 15)** to trap large impurities during loading operation.
2. The dryer was also provided with stairs and platforms with railings **(Figure 16)**.
3. The belt-pulley transmission of the furnace was provided with cover **(Figure 17)**.

Figure 14. Operator of the dryer

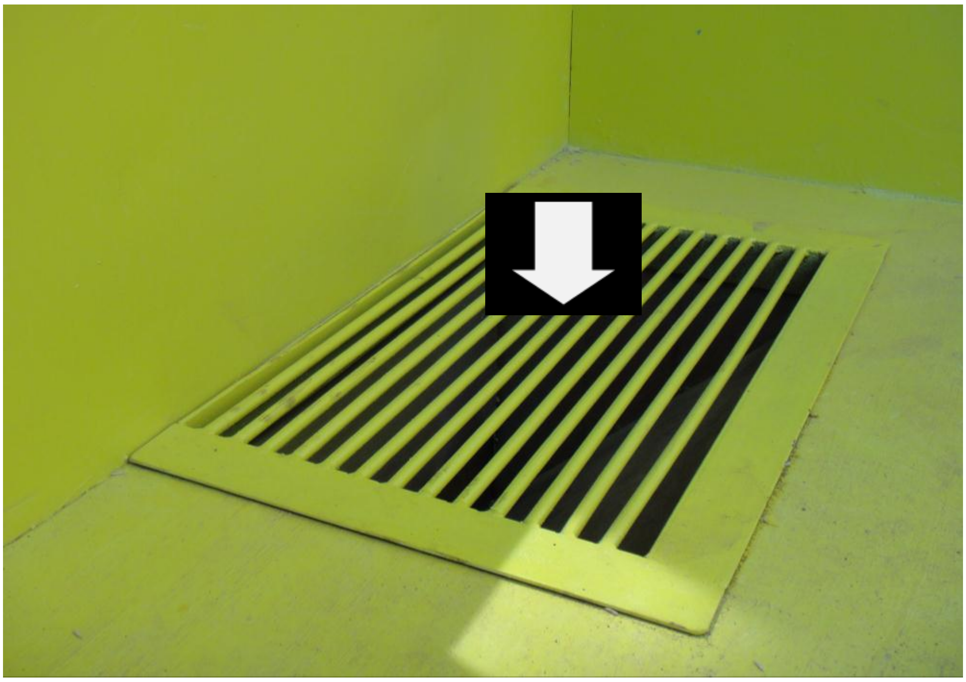

Figure 15. Grill cover of the elevator's hopper

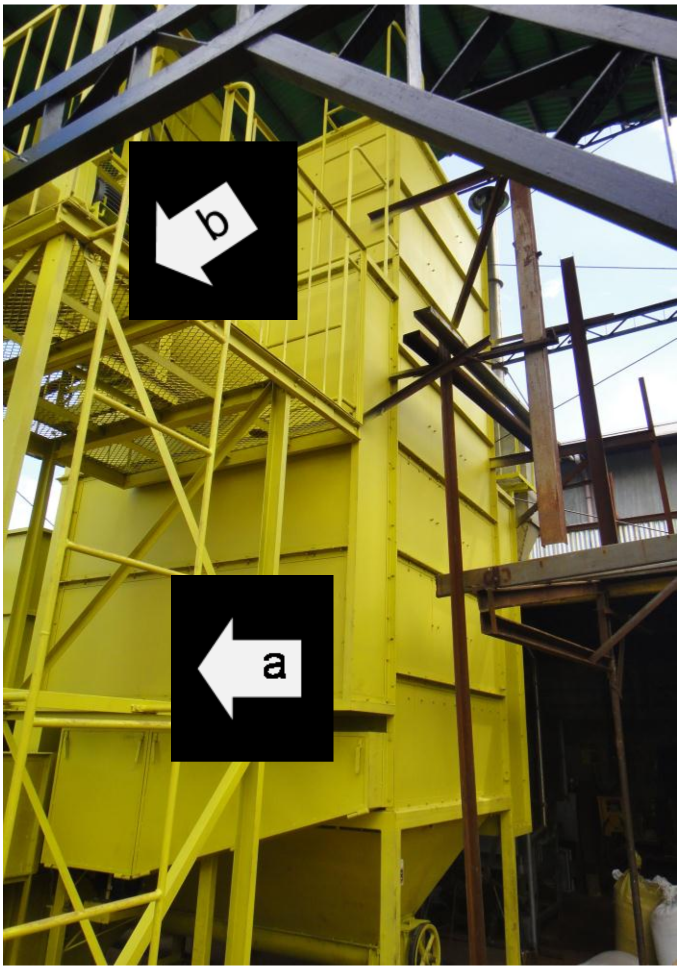

Figure 16. a) Stair
b) Platform with railing

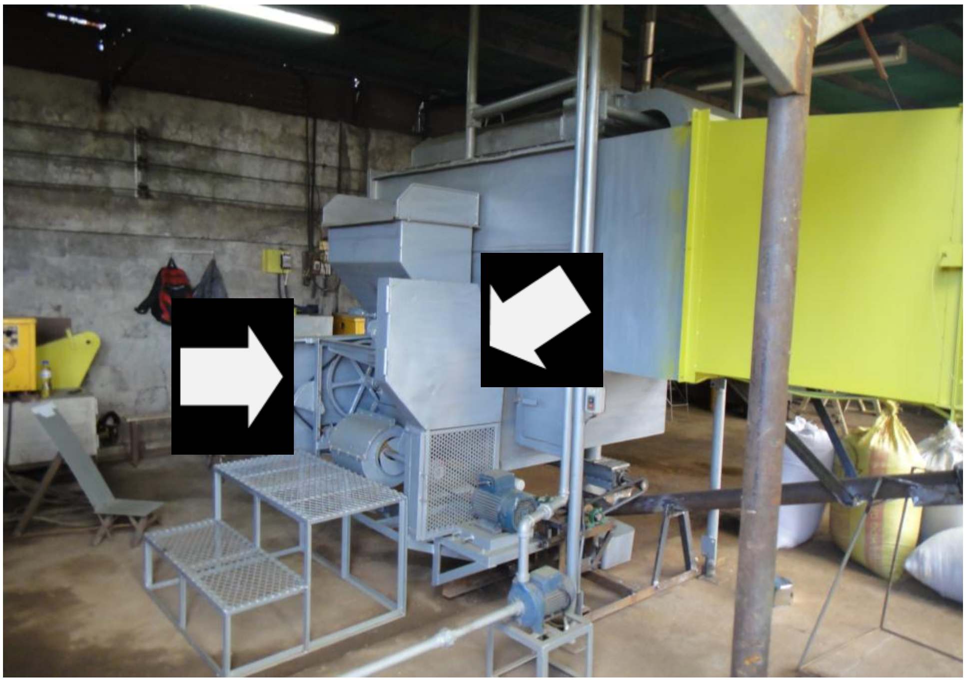

Figure 17. Belt covers

Table 1. Results of Performance Test of LAKAS KULIGLIG RMD-LK 66 Recirculating Grain Dryer

ITEMS	DATA
1. Initial Weight of Test Material (kg)	6000
2. Final Weight of Test Material (kg)	4935
3. Initial Moisture Content (%)	29.1
4. Final Moisture Content (%)	13.9
5. Actual Drying Time (h)	15.0
6. Drying Capacity (kg/h)	400.0
7. Rate of Moisture Removal (kg/h)	71.0
8. Moisture Reduction Rate (%/h)	1.02
9. Average Drying Air Temperature (^{o}C)	60.8
10. Ambient Air Temperature (^{o}C) a) Dry Bulb b) Wet Bulb	 26.9 23.9
11. Ambient Air Relative Humidity (%)	78.0
12. Dryer Exhaust Air Temperature (^{o}C) a) Dry Bulb b) Wet Bulb	 33.6 31.3
13. Exhaust Air Relative Humidity (%)	84.9
14. Grain Temperature at the Drying Section (^{o}C)	30.4
15. Fan Air Velocity (m/s)	4.35
16. Air Flow Rate (m^3/min)	75.6
17. Plenum Static Pressure (mmH_2O)	14
18. Speed of Components (rpm) a) Biomass Fuel Feed Plunger b) Fan for Supply of Combustion Air c) Discharge Ash Screw Conveyor	 14.5 944 1764
19. Furnace Fuel Consumption Rate, using rice hull (kg/h)	24.7
20. Average Power Consumption During Drying Process a) Input Power, kW b) Line Voltage, V c) Load Current, A	 7.30 221 42.1

Table 1. Results of Performance Test of LAKAS KULIGLIG RMD-LK 66 Recirculating Grain Dryer

ITEMS	DATA
21. Heating System Efficiency (%)	53.7
22. Heat Utilization (kJ/kg moisture removed)	2600
23. Drying Efficiency (%)	90.5
24. Drying System Efficiency (%)	48.5
25. Moisture Content Variation at the Termination of Drying (%)	± 0.24
26. Moisture Gradient (%)	1.10

Table 2. General information of air-dried palay samples brought to AMTEC laboratory for analysis

ITEMS	DATA
1.Variety	Mixed (C-10, C-216, C-222)
2.Source	Rizal, Nueva Ecija
3.<u>Paddy Analysis</u>	
3.1 Grain Size, brown rice	
a) Length	6.64
b) Width	2.29
c) Thickness	1.88
3.2 Grain Type	Long and slender
3.3 Paddy Bulk Density (kg/m^3)	460.5
3.4 Moisture Content (% w. b.)	14.1
3.5 Purity (%)	96.77
3.6 Cracked (%)	0.67
3.7 Immature (%)	3.08
3.8 Chalky (%)	1.52
3.9 Yellow and Fermented (%)	1.08
3.10 Red Rice (%)	2.0
4. <u>Laboratory Milling Test</u>	
4.1 Total Milling Recovery (%)	67.0
4.2 Head Rice Recovery (%)	
a) Based on Milled Rice	88.7
b) Based on Rough Rice	59.4
4.3 Broken Rice (%)	
a) Based on Milled Rice	10.7
4.4 Brewer's Rice (%)	
a) Based on Milled Rice	0.6

Table 3. General information of samples dried using the LAKAS KULIGLIG RMD-LK 66 Recirculating Grain Dryer brought to AMTEC laboratory for analysis

ITEMS	DATA
1. Variety	Mixed (C-10, C-216, C-222)
2. Source	Rizal, Nueva Ecija
3. Paddy Analysis	
3.1 Grain Size, brown rice	
a) Length	6.84
b) Width	2.48
c) Thickness	1.78
3.2 Grain Type	Long and slender
3.3 Paddy Bulk Density (kg/m^3)	477.2
3.4 Moisture Content (% w. b.)	13.9
3.5 Purity (%)	97.08
3.6 Cracked (%)	2.67
3.7 Immature (%)	8.40
3.8 Chalky (%)	7.40
3.9 Yellow and Fermented (%)	1.92
3.10 Red Rice (%)	0.08
4. Laboratory Milling Test	
4.1 Total Milling Recovery (%)	66.7
4.2 Head Rice Recovery (%)	
a) Based on Milled Rice	92.4
b) Based on Rough Rice	61.7
4.3 Broken Rice (%)	
a) Based on Milled Rice	6.9
4.4 Brewer's Rice (%)	
a) Based on Milled Rice	0.7

ANNEX 1. FORMULA USED IN THE COMPUTATION OF DRYING PARAMETERS

1. Drying Capacity (kg/h) $= \dfrac{W_1}{T_A}$

where,

W_1 = Initial weight of test material (kg)
T_A = Actual drying time (h)

2. Final Weight of Test Materials, (kg)

$$W_2 = \frac{W_1\,(100 - MC_1)}{(100 - MC_2)}$$

Where,

W_2 = Final weight of test materials (kg)
MC_1 = Initial moisture content (%)
MC_2 = Final moisture content (%)

3. Moisture Reduction Per Hour (kg/h)

$$\text{Moisture reduction per hour} = \frac{W_1 - W_2}{T_A}$$

4. Heating System Efficiency, E_{HS} (%)

$$E_{HS} = \frac{Q_S}{Q_T} \times 100$$

Where,

E_{HS} = Heating system efficiency (%)
Q_s = Heat supplied to the dryer (kJ/h)
Q_f = Heat available in the fuel (kJ/h)

To solve for Q_s:

$$Q_s = \frac{\Delta h \times Q}{v} \times 60$$

Where,

Δh = Change in enthalpy (kJ/kg d.a.)
Q = Air flow rate (m^3/min)
v = Specific volume of fresh air (m^3/kg d.a.)

To solve for Q_f:

$$Q_f = F_c \times H_{vf}$$

Where,

F_c = Fuel consumption (kg/h)
H_{vf} = Heating value of the fuel used (kJ/kg)

5. Heat utilization, Q_u (kJ/kg)

$$Q_u = \frac{Q_S \times T_A}{W_1 - W2}$$

6. Drying Efficiency (%)

$$E_D = \frac{Q_R}{Q_S} \times 100$$

Where,

E_D = Drying efficiency (%)
Q_R = Heat required to vaporize moisture inside the grain (kJ/h)

To solve for Q_R

$$Q_R = Q_v - (W_1 - W_2)/T_A$$

Where,

Q_v = Heat of vaporization of water

7. Drying System Efficiency, E_{DS} (%)

$$E_{DS} = Q_T/Q_f \times 100$$

N.B.

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